

# Kimi-K3-Neuron-IQ1S-GGUF — independent bench report

**Bench date:** 2026-08-08 · **Benchd by:** NetSphere estate (Killy)

**Hardware:** 4× RTX PRO 6000 Blackwell Workstation Edition (sm\_120, 96 GB each, 384 GB total), 52 vCPU / 639 GiB RAM, NVMe (measured 13.1 GB/s)

**Engine:** llama.cpp custom sm120/cuda13.3 build, single stream (--parallel 1)

**Launch flags (identical to our official-IQ1\_S baselines):** ctx 262144 · ub 2048 · flash-attn on · temp 1.0 / top-p 0.95 / top-k 0 / min-p 0 · reasoning\_effort=max · --reasoning-budget 16384 · --no-cache-prompt -sps 0 (K3 multi-turn KDA-state correctness flags)

## Artifact verification

- 9 shards, **307.49 GiB**, byte-sizes match the repo listing.
- GGUF header: arch kimi-k3, 93 blocks, 896 experts / 16 active / 4 shared — **full expert structure retained**, as the card claims.
- expert\_feed\_forward\_length **1536 vs 3072 official** — the pruning is exactly **50% of every expert's FFN channels**; dense FFN (33792) and embeddings (7168) untouched.
- **No nextn/MTP tensors — but the official unsloth GGUF has none either** (scanned all 14 shards). Nothing was lost to pruning on that front; MTP-based speculative decode was never available to K3 GGUFs on this engine.
- Load-time warning special\_eos\_id is not in special\_eog\_ids (tokenizer config): **benign in practice** — generation converges and stops correctly across 100+ eval items.
- Loads healthy in **21–30 s** from page cache and goes **fully GPU-resident (~355 GB incl. KV/buffers)** — the first K3 quant that fits this estate's VRAM. The 594 GB official IQ1\_S must spill ~210 GB of experts to CPU/mmap on the same hardware.

## Speed (same harness, same flags, same machine as baselines)

| metric                               | official UD-IQ1_S (594 GB, CPU-spill) | Neuron (330 GB, GPU-resident) | ×               |
|--------------------------------------|---------------------------------------|-------------------------------|-----------------|
| deep prefill @ 4K / 16K / 32K prompt | 123 / 150 / 158 tok/s                 | <b>581 / 665 / 641</b>        | <b>4.1–4.7×</b> |
| deep decode @ 4K / 16K / 32K         | 12.29 / 12.30 / 12.19                 | <b>20.21 / 20.01 / 19.79</b>  | <b>1.6×</b>     |
| short-prompt decode                  | ~11.5                                 | <b>20.3</b>                   | <b>1.8×</b>     |
| 100K-prompt prefill (deepctx replay) | 84.6 (q2kxl datum)                    | <b>423.6–424.2</b>            | ~5×             |

Config matrix (pre-registered, one variable at a time, ≥2 reps): flash-attn on/off/auto, ubatch 512/2048/4096, threads 8/52, sampler 2×2 (top\_k × min\_p). **Incumbent config optimal on every axis.** fa off cannot even allocate at 262K ctx; ub2048 = ub4096 > ub512 on prefill (664 vs 399); threads and sampling are speed-neutral. Decode is kernel-latency-bound (tiny 1536-wide experts), insensitive to every lever tried — the gain over official is residency, and it saturates there.

## Quality

**Full v2 suite** (80 scored items, seed 7, ctx-tokens 100K, deepctx 15 / agentic 15 / grounded 30 / math-hard 20):

| model                               | agent | deepctx      | grounded | math-hard | OVERALL      |
|-------------------------------------|-------|--------------|----------|-----------|--------------|
| GLM TR3 (estate resident, 90 tok/s) | 15/15 | 7/15         | 30/30    | 19/20     | 71/80        |
| <b>k3-neuron (18 tok/s)</b>         | 13/15 | <b>15/15</b> | 29/30    | 11/20     | <b>68/80</b> |

Overall statistically indistinguishable from the resident (McNemar  $p = 0.65$ ) — but the categories are violently opposed, both at  **$p = 0.0078$** :

- **deepctx 15/15, perfect at every needle depth (5/5 at  $d=0.1/0.5/0.9$ )** vs TR3's 7/15. On the hardest-5 replay (120K-token prompts): **4/5, including a PASS on contradiction-009 — an item with a 0-7/22 field record that every previously-benched model failed.**
- **math-hard 11/20** vs 19/20. Caveat in your favor: our server caps reasoning at 16,384 tokens while the TR3 run thought up to 32K, so part of this gap is budget asymmetry, not capability.

**Same-model control** (narrow-15 vs our unpruned 803 GB Q2\_K\_XL, identical server flags and budget): **9/15 vs 15/15** ( $p = 0.0312$ ). Failure anatomy on math-hard (1/5 vs 5/5):

- **3 × non-convergence** — finish=length at exactly the reasoning budget; the model reasons without landing where the unpruned quant closes in the same budget.
- **1 × digit-slip** — converged but 53401/24266 vs correct 53401/24366.
- Agentic (3/5): two failures produced **empty final answers** after completed tool loops — protocol degradation, not wrong values.
- Grounded retrieval: clean (5/5 narrow, 29/30 full).

## The budget-vs-pruning attribution (follow-up run)

The 16K reasoning cap was our server's config, not yours, so we re-ran the nine full-v2 math failures at a **32,768 reasoning budget**, then fed whatever still failed to the **official unpruned IQ1\_S at the same budget** — isolating budget from pruning from bitwidth.

- **6 of 9 rescued at 32K** (converged at 10.1–18.6K thinking tokens — past the old cap). Neuron's honest math-hard at 32K is **17/20**, and its full-v2-equivalent rises to **~74/80 — above the estate resident's 71/80**, with deep-context doing the lifting.
- **The 3 stubborn items went to the official IQ1\_S at 32K, which solved 2 — fast.** Same bitwidth, no pruning: on those items the pruning is the culprit, and the mechanism is **reasoning efficiency** — 2–5× the thinking tokens to close an argument — not a capability ceiling.
- 13\_recur fails the official IQ1\_S too (fast but wrong) while our 803 GB Q2\_K\_XL solves it — a **bitwidth casualty**, not yours.

| item             | Neuron @ 32K                    | official IQ1_S @ 32K |
|------------------|---------------------------------|----------------------|
| math_hard_03_det | FAIL — burned 32,768, no answer | PASS — 6,551 tokens  |

| item               | Neuron @ 32K                           | official IQ1_S @ 32K                  |
|--------------------|----------------------------------------|---------------------------------------|
| math_hard_12_order | FAIL — 25,083 tokens, wrong (15 vs 58) | <b>PASS — 3,518 tokens</b>            |
| math_hard_13_recur | FAIL — burned 32,768, no answer        | FAIL — 4,727 tokens, wrong (3 vs 116) |

# Verdict

The pruning method preserved attention/retrieval **perfectly** — consistent with the design (channels cut inside expert FFNs only; attention kept Q8) — and the cost is concentrated in **deep serial reasoning**, expressed mostly as non-convergence rather than wrong answers, plus occasional last-digit numeric slips. Your card's 73.7% top-1 agreement disclosure is consistent with what we measured.

**As a long-context reading/extraction body this is exceptional per-GB — it beat our 400K-context resident on deep context while being 44% smaller than your own size class.** As a math/agent brain, the unpruned Q2\_K\_XL remains clearly better at equal budget. If the method allows a selective variant (e.g., prune shallower on layers that host reasoning-heavy experts, or 60/40 instead of 50/50), the deepctx result says the retrieval side has margin to spare.

Not tested: your WikiText-2 PPL protocol (no perplexity binary in our engine image) and HumanEval (different harness). Raw artifacts available: config matrix jsonl, ladder/deep JSONs, full netterm-eval run dirs with per-item transcripts.